

A DETAILS ON THE TWO PROMPTING STRATEGIES

Assume an LLM with context window size W is used to summarize an input document D whose length $L \gg W$. We thus split D into non-overlapping chunks $c_1, c_2, \dots, c_{\lceil \frac{L}{C} \rceil}$ where $C < W$ is the length of each chunk.

A.1 HIERARCHICAL MERGING

Hierarchical merging works as follows:

1. Obtain summaries at the base level $l = 0$ by summarizing each chunk.
2. Obtain summaries for the first level $l = 1$ by prompting the LLM to merge as many consecutive level-0 summaries $s_i, s_{i+1}, \dots$ as possible¹⁷ such that the total length of the merging prompt, the selected summaries, and the prior context (if there exists a preceding summary at the same level) is less than $W - G_l$, where G_l is a hyperparameter controlling summary length that varies depending on the level l .
3. Repeat the previous step recursively until we are left with a single summary for the book.

A.2 INCREMENTAL UPDATING

Incremental updating works as follows:

1. Feed the summarization prompt into the LLM along with the first chunk c_1 to obtain a summary of the first chunk, which initializes the global summary g_1
2. Now, provide the LLM with the updating prompt, the next chunk c_2 , and the current global summary g_1 . The model is prompted to updating the global summary to g_2 with information from the current chunk.
3. Iterate through the remaining chunks. If g_i exceeds the maximum summary length G_n , call the compression prompt to compress g_i to fit within the length limit.¹⁸ See Appendix A.2.1 for more details on compression.

A.2.1 COMPRESSION

The compression step is required for incremental updating. Through our experimentation, we have observed that as the model processes a book through incremental updating, it consistently adds more information to the running summary instead of removing things. Even with an updating prompt, the summary often surpasses the target length as removing content from it is not in the model’s natural inclination. Thus, a separate prompt is needed for the model to condense the summary. However, in hierarchical summarization, condensing is not required. The merging step is less likely to run over the summary limit since it does not have to work with a pre-existing running summary. If the summaries generated during hierarchical merging go over the summary limit, simply asking the model to regenerate up to a fixed number of times would suffice.

B DATASET DETAILS

B.1 TABLE OF ALL BOOKS IN THE DATASET

See Table 3.

¹⁷Wu et al. (2021) suggest that since independent chunk-level summarization might miss vital context from earlier sections of the story, we can mitigate this effect by joining as many preceding summaries from the same level as possible. We thus implement this approach in our method.

¹⁸As the final summary often contains artifacts like “in the current segment” or “in this section”, especially with weaker models, we included an additional post-processing prompt to clean up these artifacts. See Appendix M.3.

Table 3: Title, author, genres, and publication date of all books in our dataset, sorted alphabetically.

Name	Author	Genres	Published
A Day of Fallen Night	Samantha Shannon	fantasy, fiction, lgbt	2013/02/28
A Fever in the Heartland: The Ku Klux Klan's Plot to Take Over America, and the Woman Who Stopped Them	Timothy Egan	crime, historical, non-fiction	2023/04/04
A Girl Called Samson	Amy Harmon	fiction, historical, romance	2023/04/01
A Heart That Works	Rob Delaney	contemporary, family, memoir, non-fiction	2022/11/29
A History of Burning	Janika Oza	fiction, historical	2023/05/02
A House with Good Bones	T. Kingfisher	fantasy, fiction, gothic, horror	2023/03/28
A Likely Story	Leigh McMullan Abramson	contemporary, fiction, mystery	2023/03/14
A Living Remedy: A Memoir	Nicole Chung	biography, contemporary, memoir, non-fiction	2023/04/04
Age of Vice	Deepthi Kapoor	crime, fiction, thriller	2023/01/03
All the Dangerous Things	Stacey Willingham	fiction, thriller	2023/01/10
Anders & Santi Were Here	Jonny Garza Villa	contemporary, fiction, lgbt, romance	2023/05/02
Atalanta	Jennifer Saint	fantasy, historical, mythology, retelling	2023/04/11
Black Cake	Charmaine Wilkerson	contemporary, fiction, historical, mystery	2022/02/01
Camp Zero	Michelle Min Sterling	fantasy, fiction, scifi	2023/04/04
Catfish Rolling	Clara Kunagai	fantasy, fiction, lgbt, mythology, scifi	2023/03/02
Central Places	Delia Cai	contemporary, fiction	2023/01/31
Chain-Gang All-Stars	Nana Kwame Adjei-Brenyah	fantasy, fiction, lgbt, scifi	2023/05/02
City Under One Roof	Iris Yamashita	crime, fiction, mystery, thriller	2023/01/10
Clytemnestra	Costanza Casati	fantasy, fiction, historical, mythology	2023/05/02
Deep as the Sky, Red as the Sea	Rita Chang-Eppig	fantasy, fiction, historical	2023/05/30
Did You Hear About Kitty Karr?	Crystal Smith Paul	fiction, historical	2023/05/02
Divine Rivals	Rebecca Ross	fantasy, historical, romance	2023/04/04
Drowning	T. J. Newman	fiction, mystery, thriller	2023/05/30
Emily Wilde's Encyclopaedia of Faeries	Heather Fawcett	fantasy, fiction, historical, romance	2023/01/10
Flowerheart	Catherine Bakewell	fantasy, fiction, romance	2023/03/14
Ghost Music	An Yu	contemporary, fantasy, fiction	2022/01/01
Good Night, Irene	Luis Alberto Urrea	fiction, historical	2023/05/30
Greek Lessons	Han Kang	contemporary, fiction, romance	2023/04/18
Greyhound Fair	Francesca Zappia	fantasy, horror, mystery, retelling	2023/03/28
Gwen and Art Are Not in Love	Lex Croucher	fantasy, fiction, historical, lgbt, romance	2023/05/11
Happy Place	Emily Henry	contemporary, fiction, romance	2023/04/25
Heart of the Sun Warrior	Sue Lynn Tan	fantasy, fiction, mythology, retelling	2022/11/10
Homecoming	Kate Morton	fiction, historical, mystery, thriller	2023/04/13
Honeybees and Distant Thunder	Riku Onda	contemporary, fiction	2023/05/02
How to Turn into a Bird	Maria José Ferrada	contemporary, fiction	2021/12/06
I Have Some Questions for You	Rebecca Makkai	contemporary, fiction, mystery, thriller	2023/02/21
I Want to Die But I Want to Eat Tteokpokki	Beck Sehee	memoir, non-fiction	2022/11/01
In the Lives of Puppets	T. J. Klune	fantasy, fiction, lgbt, romance, scifi	2023/04/25
Into the Light	Mark Oshiro	contemporary, fiction, lgbt, mystery, thriller	2023/03/28
Isba, Unscripted	Sajni Patel	contemporary, fiction, romance	2023/02/14
Jana Goes Wild	Farah Heron	comedy, contemporary, fiction, romance	2023/05/02
Lady Tan's Circle of Women	Lisa See	fiction, historical	2023/06/06
Lies We Sing to the Sea	Sarah Underwood	fantasy, lgbt, mythology, retelling	2023/03/30
Lone Women	Victor LaValle	fantasy, fiction, historical, horror, mystery	2023/03/28
Lunar Love	Lauren Kung Jensen	contemporary, fiction, romance	2023/01/10
Maame	Jessica George	contemporary, family, fiction	2023/01/31
Meet Me at the Lake	Carley Fortune	contemporary, fiction, romance	2023/05/02
Natural Beauty	Lang Ling Huang	contemporary, fiction, horror, lgbt	2023/04/04
Paper Names	Susie Luo	contemporary, fiction, historical	2023/05/02
Pathogenesis: A History of the World in Eight Plagues	Jonathan Kennedy	historical, non-fiction	2023/04/18
Scattered All Over the Earth	Yoko Tawada	dystopian, fiction, scifi	2022/03/01
Secretly Yours	Tessa Bailey	contemporary, romance	2023/02/07
Seven Faceless Saints	M. K. Lobb	fantasy, fiction, lgbt, mystery, romance	2023/02/07
She Is a Haunting	Trang Thanh Tran	fantasy, fiction, gothic, horror, lgbt	2023/02/28
Some Desperate Glory	Emily Tesh	fantasy, fiction, lgbt, scifi	2023/04/11
Song of Silver, Flame Like Night	Amelie Wen Zhao	fantasy, fiction, mythology, romance	2023/01/03
Spare	Prince Harry	biography, memoir	2023/01/10
Spice Road	Maiya Ibrahim	fantasy, fiction, mythology, romance	2023/01/24
The Adventures of Amina al-Sirafi	Shannon Chakraborty	adventure, fantasy, fiction, historical	2023/02/28
The Bandit Queens	Patrin Sheriff	contemporary, fiction, mystery, thriller	2023/01/03
The Book of Everlasting Things	Aanchal Malhotra	fiction, historical, romance	2022/12/27
The Collected Regrets of Clover	Mikki Brammer	contemporary, fiction, romance	2023/05/02
The Covenant of Water	Abraham Verghese	fiction, historical	2023/05/02
The Faraway World	Patricia Engel	fiction, short stories	2023/01/24
The Ferryman	Justin Cronin	fantasy, fiction, scifi, thriller	2023/05/02
The First Bright Thing	J. R. Dawson	fantasy, fiction, historical, lgbt	2023/06/13
The Girl Who Fell Beneath the Sea	Axie Oh	fantasy, fiction, mythology, retelling, romance	2022/02/22
The Golden Doves	Martha Hall Kelly	fiction, historical	2023/04/18
The Half Moon	Mary Beth Keane	contemporary, fiction	2023/05/02
The Haunting of Alejandra	V. Castro	fantasy, fiction, historical, horror, mythology, retelling	2023/04/18
The House Is on Fire	Rachel Beanland	fiction, historical, mystery	2023/04/04
The Last Pomegranate Tree	Bachtyar Ali	fantasy, fiction, historical	2023/01/03
The Last Tale of the Flower Bride	Roshani Chokshi	fantasy, fiction, gothic	2023/02/14
The Marriage Portrait	Maggie O'Farrell	fiction, historical	2022/09/06
The Mimicking of Known Successes	Malika Older	fiction, mystery, scifi	2023/03/07
The Night Travelers	Armando Lucas Correa	fiction, historical	2023/01/10
The Stolen Heir	Holly Black	fantasy, romance	2023/01/03
The Survivalists	Kashana Cauley	comedy, contemporary, fiction, life, romance	2023/01/10
The True Love Experiment	Christina Lauren (duo)	contemporary, romance	2023/05/16
The Vibrant Years	Sonali Dev	contemporary, fiction, romance	2022/12/01
The Wager: A Tale of Shipwreck, Mutiny and Murder	David Grann	adventure, crime, historical, mystery, non-fiction	2023/04/18
The Wicked Bargain	Gabe Cole Novoa	fantasy, fiction, historical, lgbt	2023/02/28
The Wishing Game	Meg Shaffer	contemporary, fantasy, fiction, mystery, romance	2023/05/30
The Words That Remain	Stenio Gardel	contemporary, fiction, lgbt, romance	2023/01/16
The Writing Retreat	Julia Bartz	fiction, horror, mystery, thriller	2023/02/21
Things I Wish I Told My Mother	Susan Patterson, Susan DiLallo, James Patterson	contemporary, family, fiction, romance	2023/04/10
Thorne Princess	L. J. Shen	contemporary, fiction, romance	2023/01/04
Tomorrow, and Tomorrow, and Tomorrow	Gabrielle Zevin	fiction, gaming, romance	2022/07/05
Tress of the Emerald Sea	Brandon Sanderson	adventure, fantasy, fiction, romance	2023/01/10
Unselkie	Ivelisse Housman	fantasy, fiction	2023/01/03
Un tethered Sky	Fonda Lee	fantasy, fiction, scifi	2023/04/11
Vera Wong's Unsolicited Advice for Murderers	Jesse Q. Sutanto	contemporary, fiction, mystery, thriller	2023/03/14
Victory City	Sulman Rushdie	fantasy, fiction, historical, mythology	2023/02/07
Walking Practice	Dolki Min	fantasy, fiction, horror, lgbt, scifi	2023/03/14
We Don't Swim Here	Vincent Tirado	fiction, horror, lgbt, mystery, thriller	2023/05/02
What Happens Next	Christina Suzann Nelson	contemporary, family, fiction, mystery	2023/01/17
While Time Remains: A North Korean Defector's Search for Freedom in America	Yeonmi Park	biography, historical, memoir, non-fiction	2023/02/14
Witch King	Martha Wells	fantasy, fiction, scifi	2023/05/30
Wrong Place Wrong Time	Gillian McAllister	fiction, mystery, thriller	2022/05/12
Yellowface	R. F. Kuang	contemporary, fiction, mystery, thriller	2023/05/16

C COARSE-GRAINED HUMAN EVALUATION

Table 4 shows detailed results of the coarse-grained human evaluation as discussed in Section 5.

Table 4: Results from coarse-grained human evaluation. Annotators compare 100 pairs of GPT-4 generated incremental and hierarchical summaries and judge (1) overall preference; (2) level of detail; (3) structure and pacing; (4) logic and understandability of each summary.

Preference	Incremental	Hierarchical	Tie
Overall	44	54	2
Detail	83	11	6
Structure	35	59	6
Logic	38	53	9

D MORE EXPERIMENT DETAILS

We use the LLaMA-2-7B-Instruct checkpoint¹⁹ fine-tuned on long-context summarization (BookSum). For Mixtral, we used the `mistralai/Mixtral-8x7B-Instruct-v0.1` checkpoint hosted by the Together API. For closed-source models, we use the `gpt-4 2023-03-15` and `gpt-3.5-turbo-0301` checkpoints on Microsoft Azure. Anthropic unfortunately does not disclose checkpoint information, but our summaries were all obtained via their Claude 2 API in September 2023.

In our experiments, LLaMA 2 uses a context window size of 4096 tokens, while other models leverage their full context window. While generating LLaMA 2 hierarchical summaries, we have to truncate the results at the final punctuation mark. If not, the model would be stuck at the regeneration phase, as it does not follow the given word limit at all. In addition, for LLaMA 2 summaries, we do not apply post-processing as described in Appendix D.1, because it would significantly alter the structure of the LLaMA 2 summaries, thereby enhancing their coherence. Instead, we post-process the summaries using a standard string matching approach to get rid of sentences copied from the prompt. Without this step, we cannot evaluate them with GPT-4, since GPT-4 would treat these prompt artifacts as instructions rather than sentences to annotate.

D.1 DATA PROCESSING DETAILS

In order to preserve all visual separators within the text, we extract all text elements from the epub files without further automatic processing. As a result, sometimes content from non-narrative sections would appear in the generated summaries. We apply simple post-processing by prompting GPT-4 to remove information coming from non-narrative sections of the book. The prompt can be found in Appendix M.3.

E EXPERIMENTS WITH SQUALITY

To investigate the effect of incremental updating and hierarchical merging on summary coherence, we evaluated GPT-4 on the validation set of the SQUALITY dataset, which contains sci-fi stories that are 4000-6000 words long. Table 5 shows the ROUGE-L scores. The baseline setting is where the model summarizes the stories in one go, truncating the stories whenever they exceed the model’s context window. Using incremental updating lowers ROUGE-L by a small amount, indicating that baseline summaries have slightly more overlap with the provided reference summaries. We will update the BOOOOKSCORE of these summaries in the next version of the paper.

¹⁹<https://github.com/togethercomputer/Llama-2-7B-32K-Instruct>